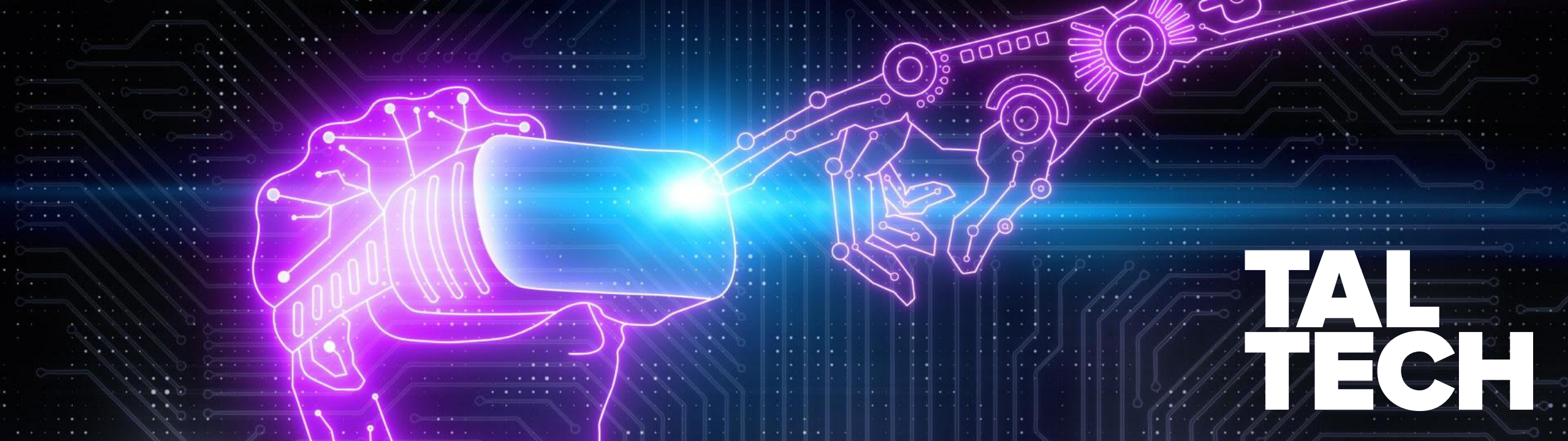




**TAL
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ITE4110: INFORMATION SOCIETY PRINCIPLES: TOWARDS E-GOVERNANCE

DR. ERIC JACKSON, JOSEPHINE LUSI

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E-GOVERNANCE TECHNOLOGIES AND SERVICES MASTER'S PROGRAM

QUICK COURSE ROUNDUP

- **Today's Agenda**

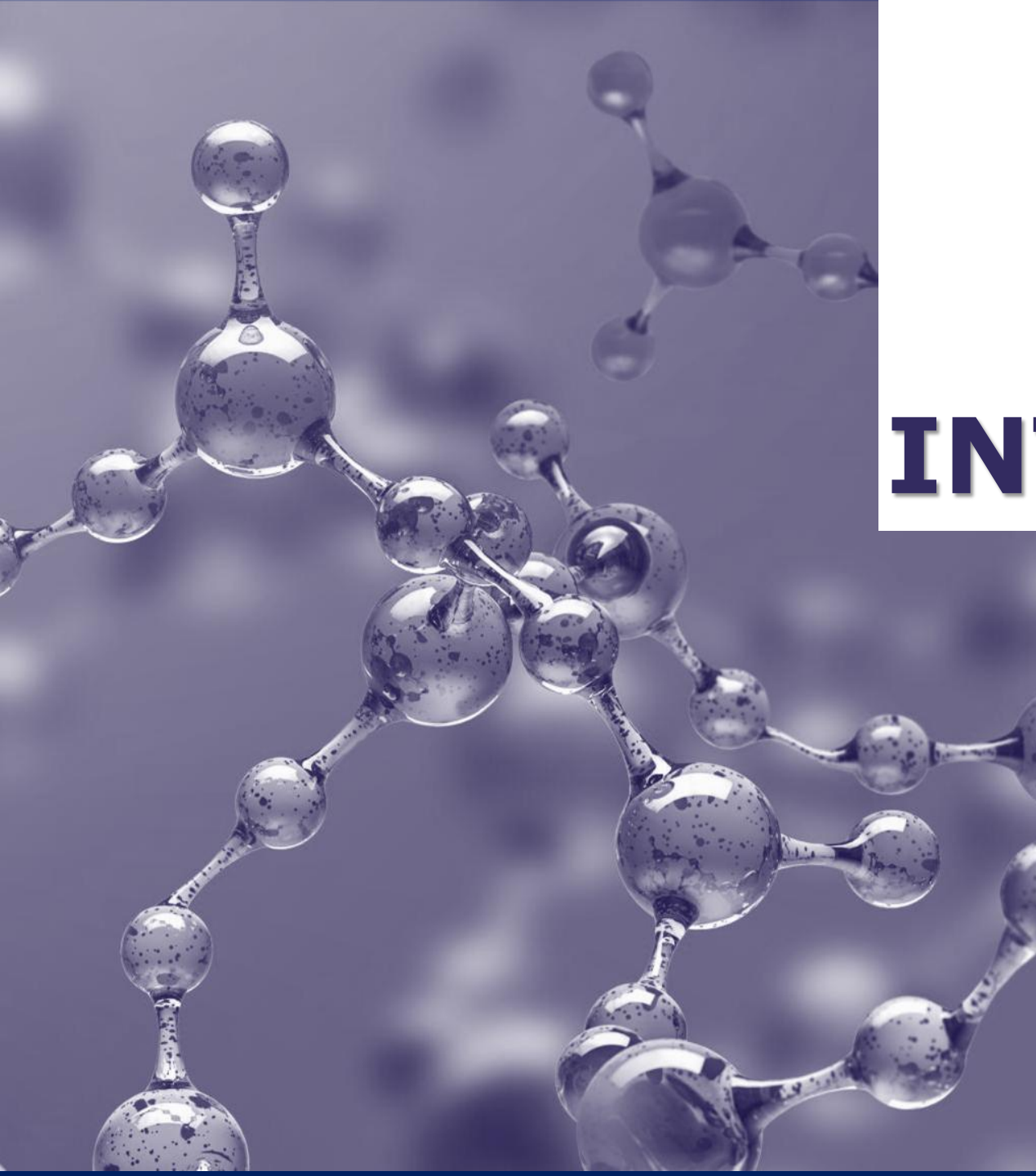
- E-Identity and Digital Signature: **Yuliia Krevchenko, e-ID Expert**
- Interoperability – **Me**
- Guest Lecturer: **Arvo Ott, e-Governance Academy Founder**

- **Task 1 Feedback**

- Will be provided by the end of the week
- Due to Moodle difficulties, the deadline was extended to Sunday of last week.

- **October 2nd Class session**

- Preliminary Case Study Overview
- Case Bank in Moodle
- Guest Lecture, John O'Duinn – Federal Government of Ireland/USA will talk about Change Management from practitioner perspective



INTEROPERABILITY

NEW EUROPEAN INTEROPERABILITY FRAMEWORK (2017)

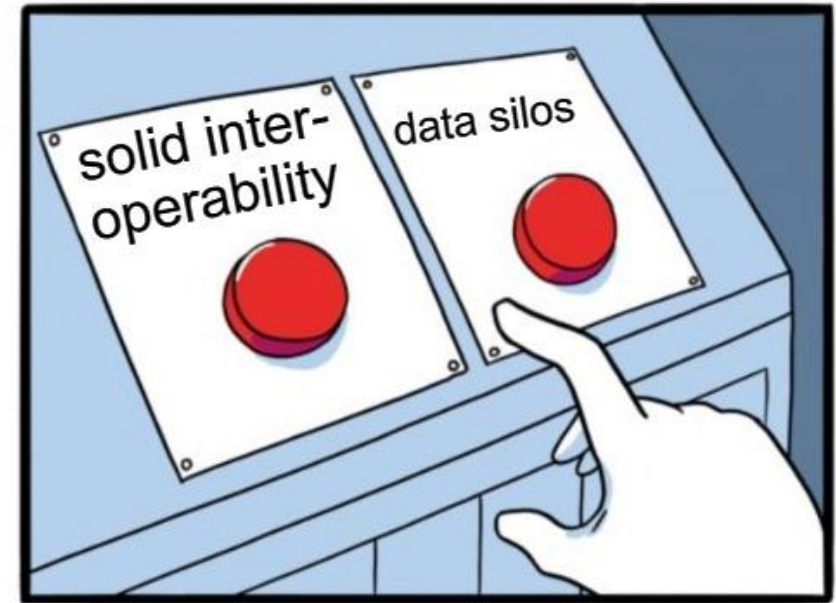
Define interoperability as: “the ability of organizations to interact towards mutually beneficial goals, involving the sharing of information and knowledge between these organizations, through the business processes they support, by means of the exchange of data between their ICT systems“

GROUP REFLECTION

- In your professional life, have you ever experienced a situation where a lack of interoperability hindered your work? Or other aspects?
- Have you seen instances where (cross-border) interoperability has facilitated your work?

INTEROPERABILITY IN BRIEF

- Public sector has a natural predilection to siloes organizationally, and with data as well.
- Interoperability is the core engine of service provision, data-driven decision-making, and innovation
- How can two people speaking different languages effectively communicate with one another?
- Organizations need to speak the same language technically and semantically...and more!
- Context matters a lot:
 - What is the purpose of exchanging data?
 - ✓ Provide e-services (G2G, G2B, G2C)
 - ✓ Communication
 - ✓ Local, state, regional, national, cross-border?



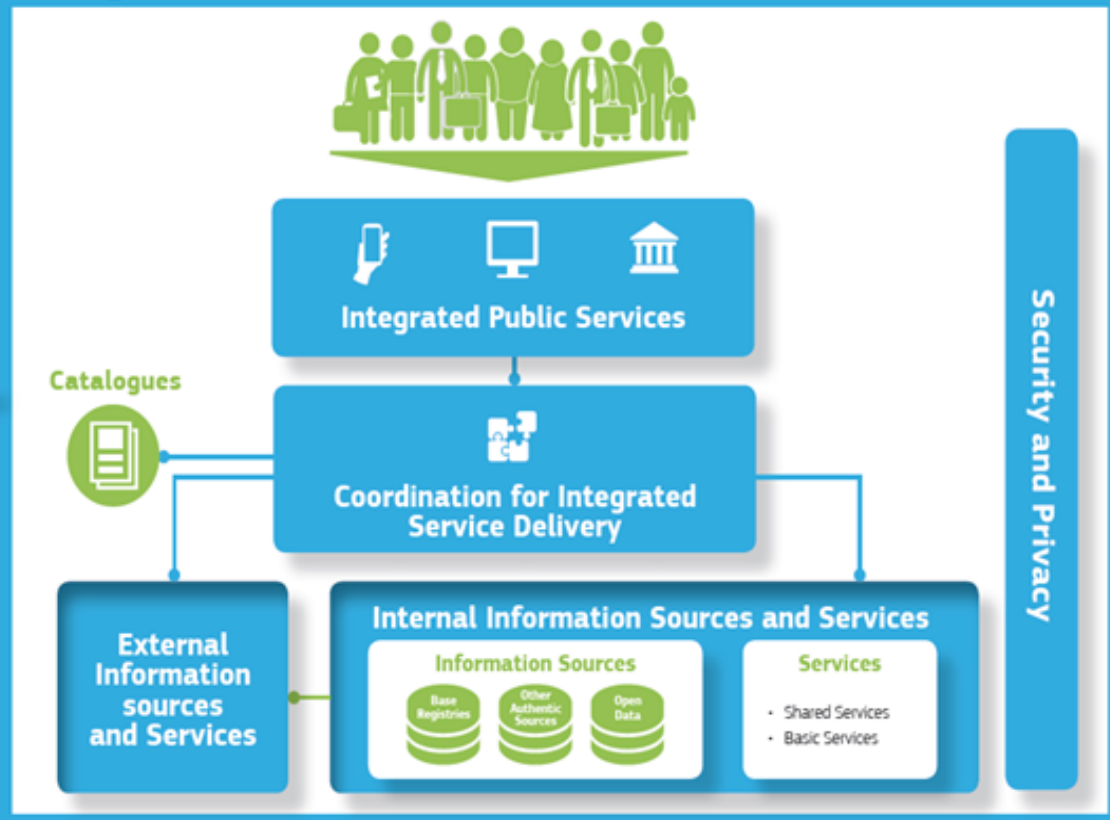
NEW EUROPEAN INTEROPERABILITY FRAMEWORK (2017)

EIF Conceptual Model

Interoperability Governance



Integrated Public Service Governance



Interoperability Principles

LEGAL LAYER



“Legal interoperability is about ensuring that organisations operating under different legal frameworks, policies and strategies are able to work together”

- **27 member-states** = different national legal frameworks, policies and strategies
- Also applies inter-organizationally in regard to policies/strategies
- Leads to **legal friction**:
 - # Legal requirements for using a single delivery method for e-services or the use of a specific digital technology by law
 - # Contradictory requirements for the same or similar business processes
 - # E-Identity and Digital Signature laws/policies
- **Cross-organizational** interoperability agreements (Internal or Cross-border)

ORGANIZATIONAL LAYER



Organizations must align their business processes, responsibilities, and expectations to achieve commonly agreed and mutually beneficial goals and document them.

- **Recommendation 28:** Document your business processes using commonly accepted modelling techniques and agree on how these processes should be aligned to deliver a European public service.
- Clearly define the relationship between service providers and service consumers, e.g., MOUs, Service Level Agreements (SLAs), API specifications, etc.
- Organizational leadership and a willingness to collaborate with other organizations

SEMANTIC LAYER



The precise format and meaning of exchanged data and information is preserved and understood throughout exchanges between parties

- “What is sent is understood”
- Cultural, linguistic, and organizational differences in the structuring and definition of data makes it difficult for interoperability to have value
- Different types of semantic tools:
 - ❖ Data glossaries, taxonomies,
 - ❖ Meta-data and Resource Description Framework
 - ❖ Shared vocabularies and reusable data models/structures
 - DCAT-AP (Data Catalogue Application Profile for European public sector administrations)
 - Data model created by the World Wide Web Consortium (W3C)

CORE ELEMENTS OF INTEROPERABILITY



Data Sharing Infrastructure

Data Exchange Systems or Data Intermediation Systems



Trust mechanisms (Authentication and Verification)

Public Key Infrastructure and Digital Signature



Standardization of data and messaging/exchange processes

REST, SOAP OpenAPIs



Data Governance (Confidentiality, Integrity and Availability)



Legal agreements regarding data accessibility, type, standards, proportionality, liability, and cooperation

WHY DO APIS MATTER FOR INTEROPERABILITY?

APIs (Application Programming Interfaces) are like contracts that allow different systems to talk to each other.

- In government, they make it possible for *tax systems, registries, healthcare platforms, and municipal services* to securely exchange data.
- Without APIs, agencies often rely on **manual file transfers or duplicating data**, which is inefficient and error-prone.

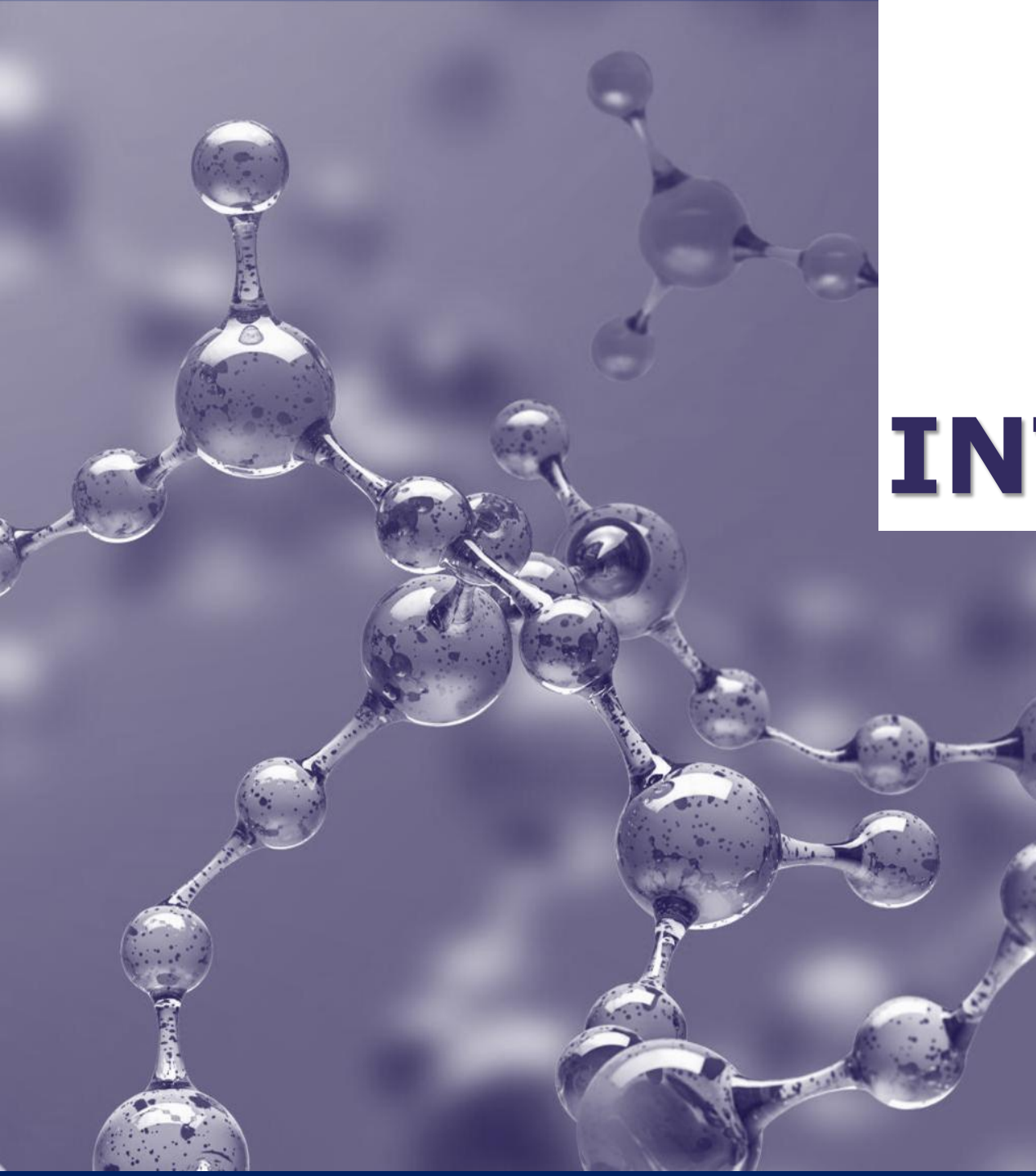
SOAP vs. REST: Two main standards in public sector systems

SOAP (Simple Object Access Protocol)

- Older, XML-based standard, widely used in many legacy government systems.
- Strong on **security (WS-Security)** and **transaction reliability**
 - Useful in contexts like tax declarations, health records, and registries.
- More rigid: requires strict contracts, which is good for critical services but slower to adapt.

REST (Representational State Transfer)

- Newer, lightweight, web-based approach (uses JSON/HTTP).
- Easier for developers, integrates well with mobile and cloud services.
- Enables **scalability and openness** – often used when governments want to open data (e.g., transport timetables, business registries, geospatial data).



INTEROPERABILITY IN ESTONIA

INTEROPERABILITY IN ESTONIA



“Once only principle”: an e-government concept that aims to ensure that citizens, institutions, and companies only have to provide certain standard information to the authorities and administrations once. (Wiki)

- Comes from a citizen-centric e-government perspective
- By incorporating data protection regulations and the explicit consent of the users, the public administration is allowed to re-use and exchange the data with each other. (Wiki)
- There is **no one central database** for storing data in Estonia.
- It is **not allowed to have duplicates of data** in Estonian ministry databases
- This ensures accountability and negates accumulation of too much power for one ministry

Interoperability in Estonia (x-tee)

- X-Tee Data Exchange Layer
- Enables internal, national, and cross-border interoperability
- Developed in Estonia in 2001, under an open-source MIT license
- Used by 600+ Estonian public and private sector authorities for data exchange
- Estonian Information Systems Authority is the operator of the X-Tee
- Codebase/community developed and maintained by Nordic Institute for Interoperable Solutions (NIIS)

Nordic Institute for Interoperable Solutions (NIIS)

- Mission - To ensure the development and strategic management of the X-Road and other cross-border components for eGovernment infrastructure
- Considered the software vendor of X-Road and the community has a say over X-Road improvements
- Can be used for internal, national, domain-based, or cross-border data exchange
- Current members: Estonia, Finland (Suomi.fi), Iceland (Straumurinn)
- There are associate partners as well: Faroe Islands, Ukraine, etc.
- Each member has their own X-Road instantiation with federative capabilities
- <https://github.com/nordic-institute/X-Road>

X-tee core components

- Architecturally distributed, with some centralized governance; however, there is no single point of failure (fault tolerance)
- One-to-one communication between service providers and service consumers (Synchronous)
- Started as a Private-Public-Academic Partnership with a consortium who won the tender for building such a system in the late 90s/early 2000s
- DATA IS KING – data is digitally signed and encrypted through security servers and travels over the public internet
- No need for expensive VPN, networking applications
- Cybernetica has a proprietary data exchange layer called UXP, modeled after the X-Road

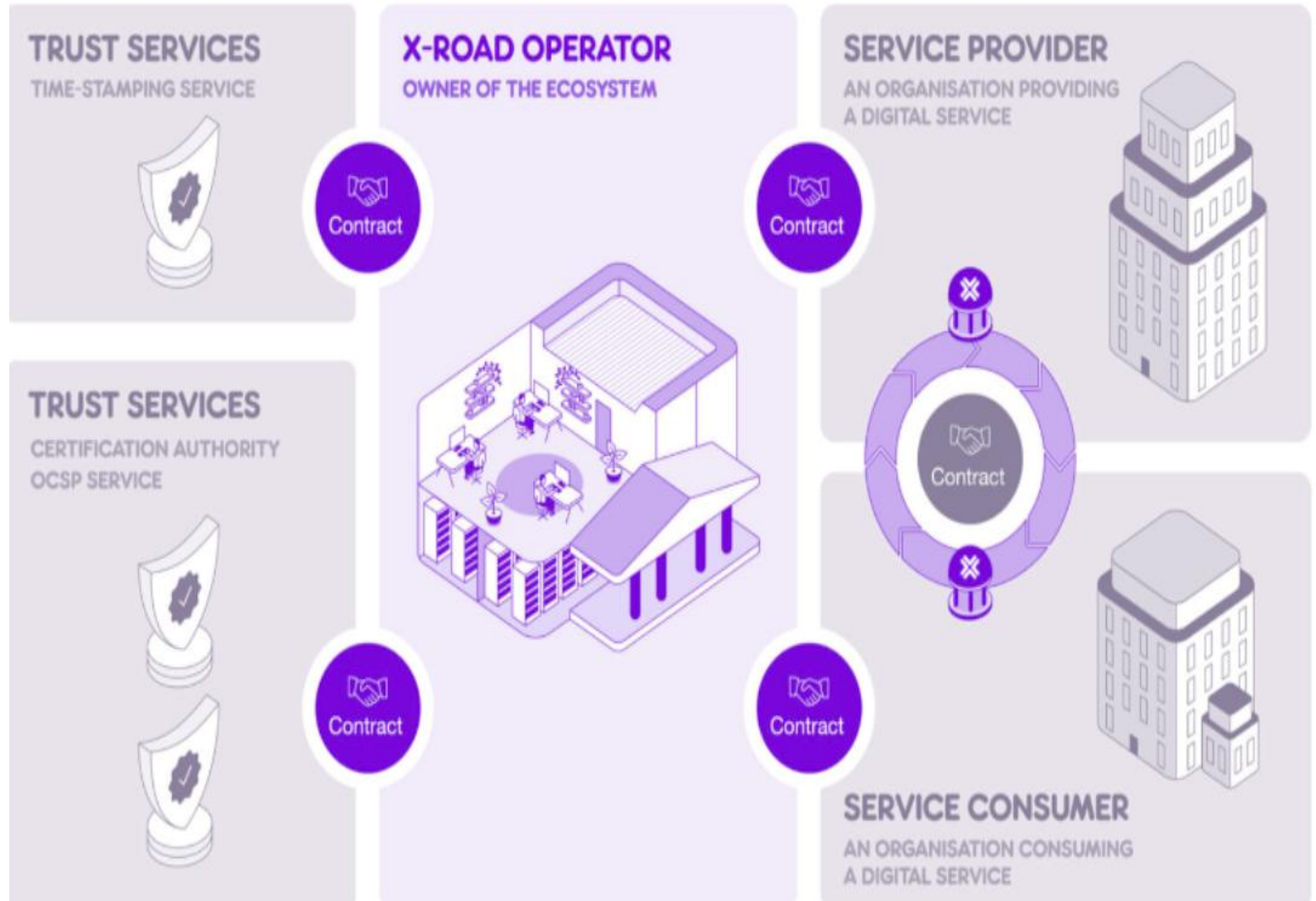
1.X-Road Operator — defines regulations, on-boards new members, operates central components

2.Members — organizations that produce and/or consume services in the ecosystem. These can be public, private, non-profit. To become a Member one must undergo an onboarding process + have technical access (security server etc.)

3.Trust Service Providers — e.g. Certification Authorities, Time-Stamping Authorities etc.

4. Technology Partners —companies offering commercial support / consulting, deployment, extensions etc. Not “members” in the sense of service producers/consumers, but officially recognized partners.

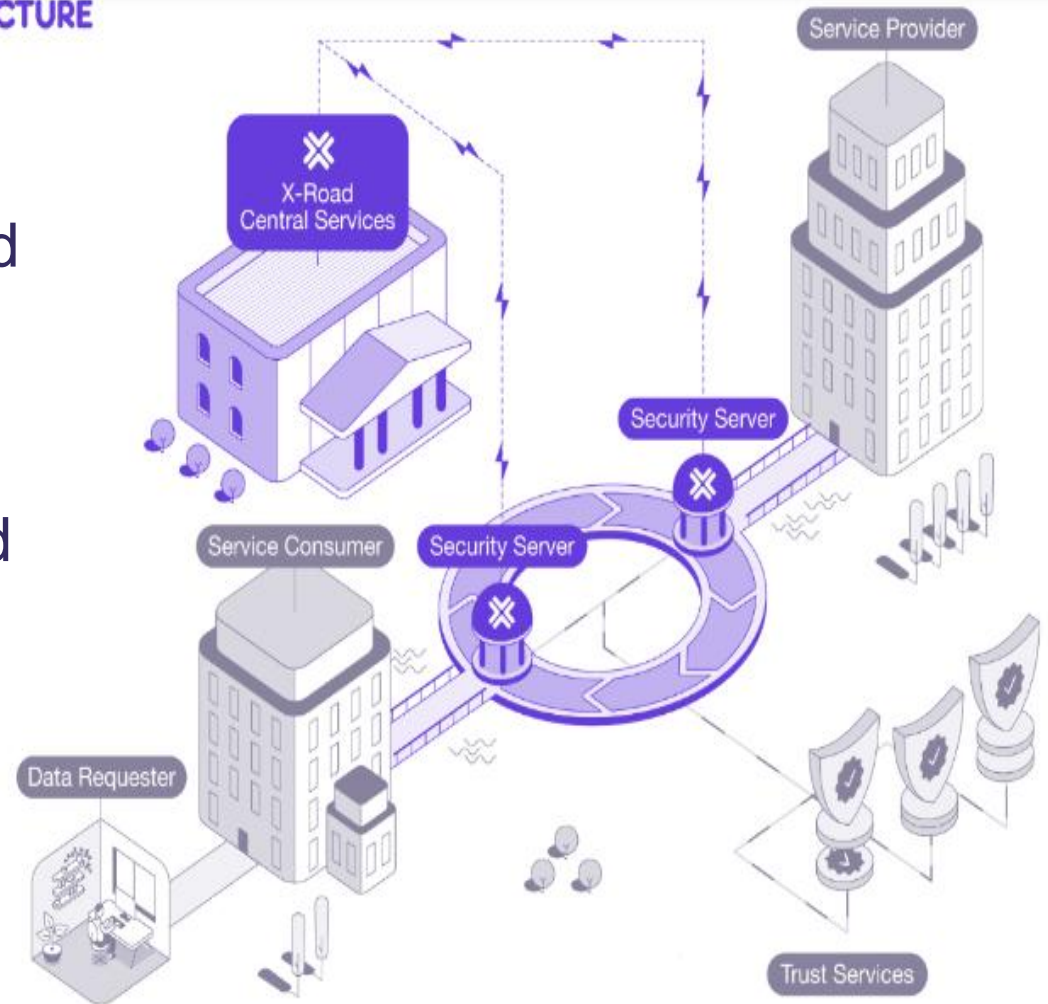
X-ROAD ECOSYSTEM



X-TEE FIVE CORE COMPONENTS (CENTRAL SERVICES)

X-ROAD ARCHITECTURE

- **Central Server/Configuration Proxy (X-Road Central Services)**
 - Central Server contains the registry of X-Road members and their Security Servers
 - Central Server contains the security policy of the X-Road instance that includes a list of trusted certification authorities, a list of trusted time-stamping authorities, and configuration parameters
 - The X-Road Operator (RIA) is responsible for operating the Central Server.
 - The Configuration Proxy can be used to increase system availability by creating an additional configuration source and reduce the load on the Central Server.

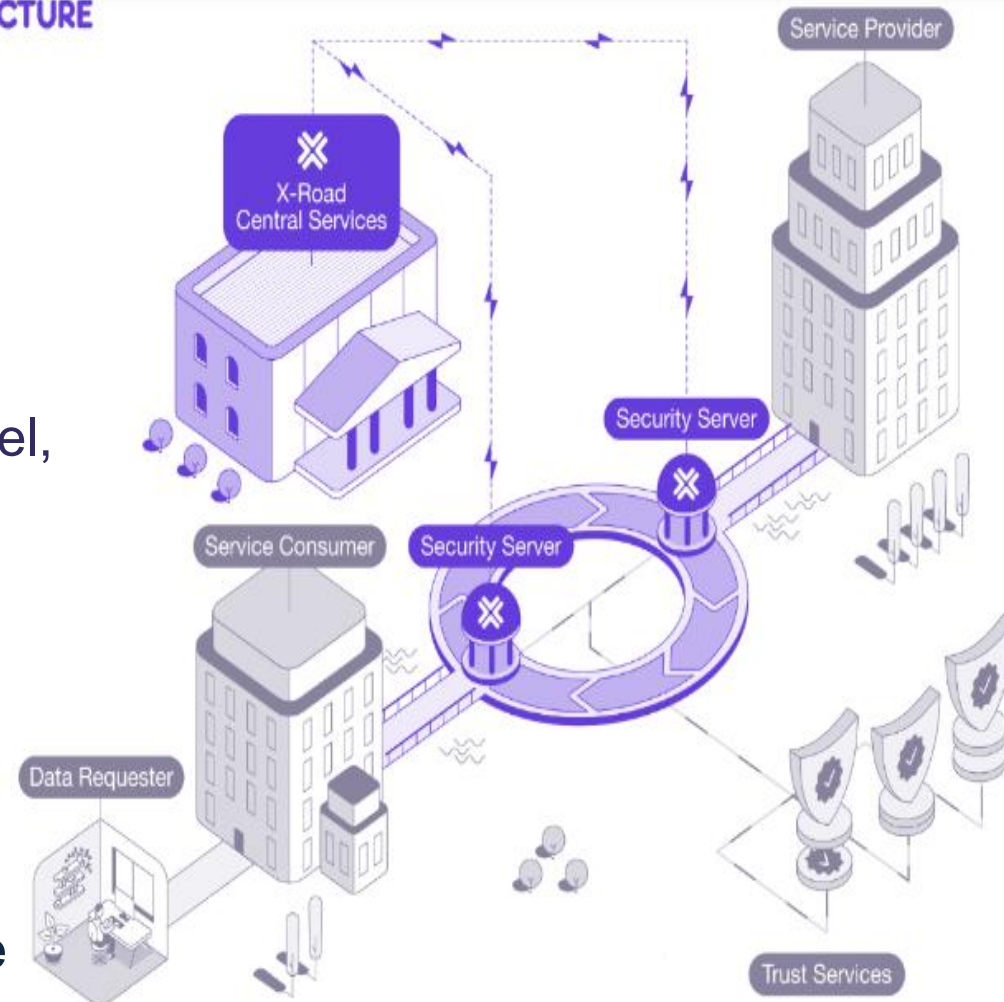


X-TEE FIVE CORE COMPONENTS (SECURITY SERVER)

- **Security Server**

- Security Server is the entry point to X-Road, and it is required for both producing and consuming services via X-Road.
- Encapsulates the security aspects of the X-Road infrastructure: managing keys for signing and authentication, sending messages over a secure channel, creating the proof value for messages with digital signatures, time-stamping and logging.
- The Security Server downloads the global configuration from the Central Server regularly and uses a local copy while processing messages.
- The Security Server manages two types of keys: Authentication keys and signing keys
- A trusted certification authority issues certificates for the keys

X-ROAD ARCHITECTURE



X-TEE FIVE CORE COMPONENTS (INFO SYSTEMS/TIME-STAMPING)

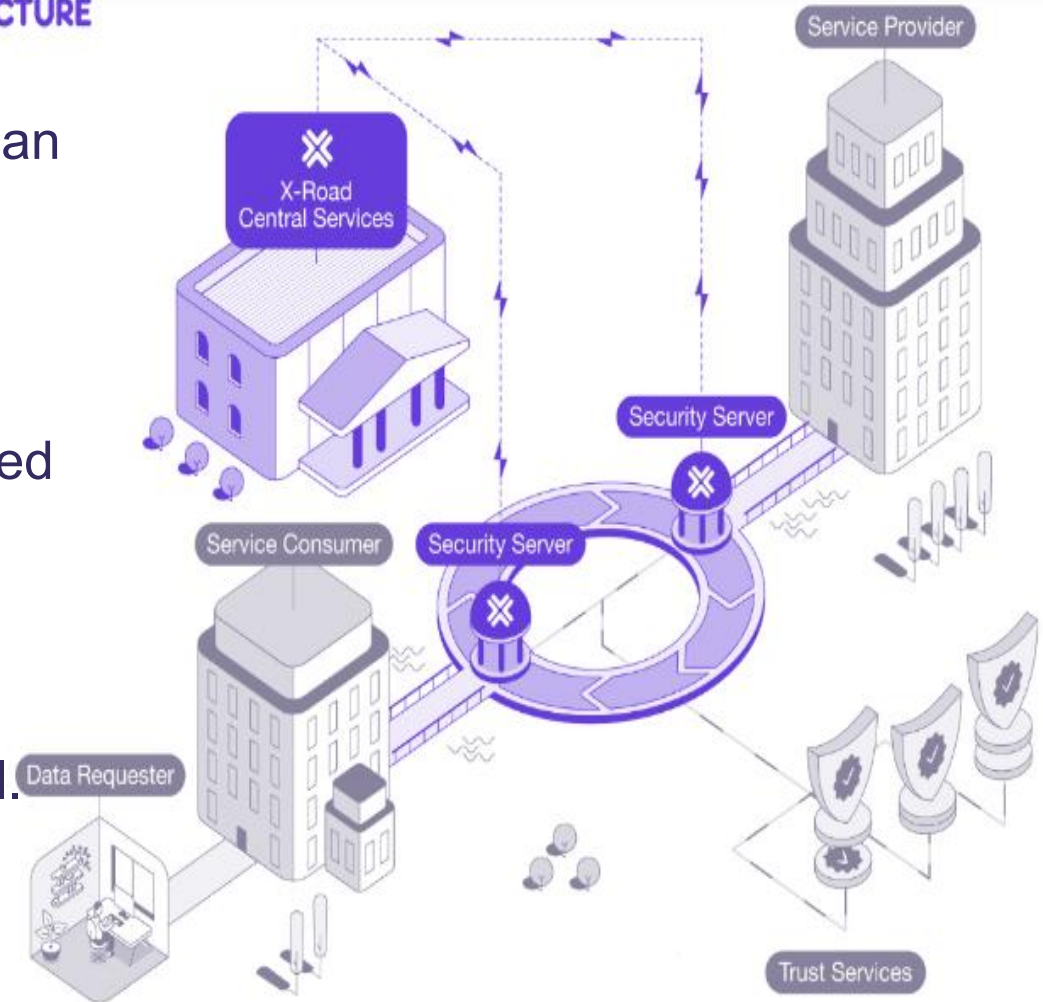
- **Information System**

- The Information System produces and/or consumes services via X-Road and is owned by an X-Road member
- Service provider implements REST/SOAP APIs and it is available via X-Road for service consumers
- Service descriptions of REST services are defined using OpenAPI3 specification, and service descriptions of SOAP services are defined using WSDL
- Service consumers can download service descriptions using the X-Road metadata protocol.

- **Time-stamping Authority**

- All the messages sent via X-Road are time-stamped and logged by the Security Server.
- TSA provides a time-stamping service that the Security Server uses for time-stamping all the incoming/outgoing requests/responses

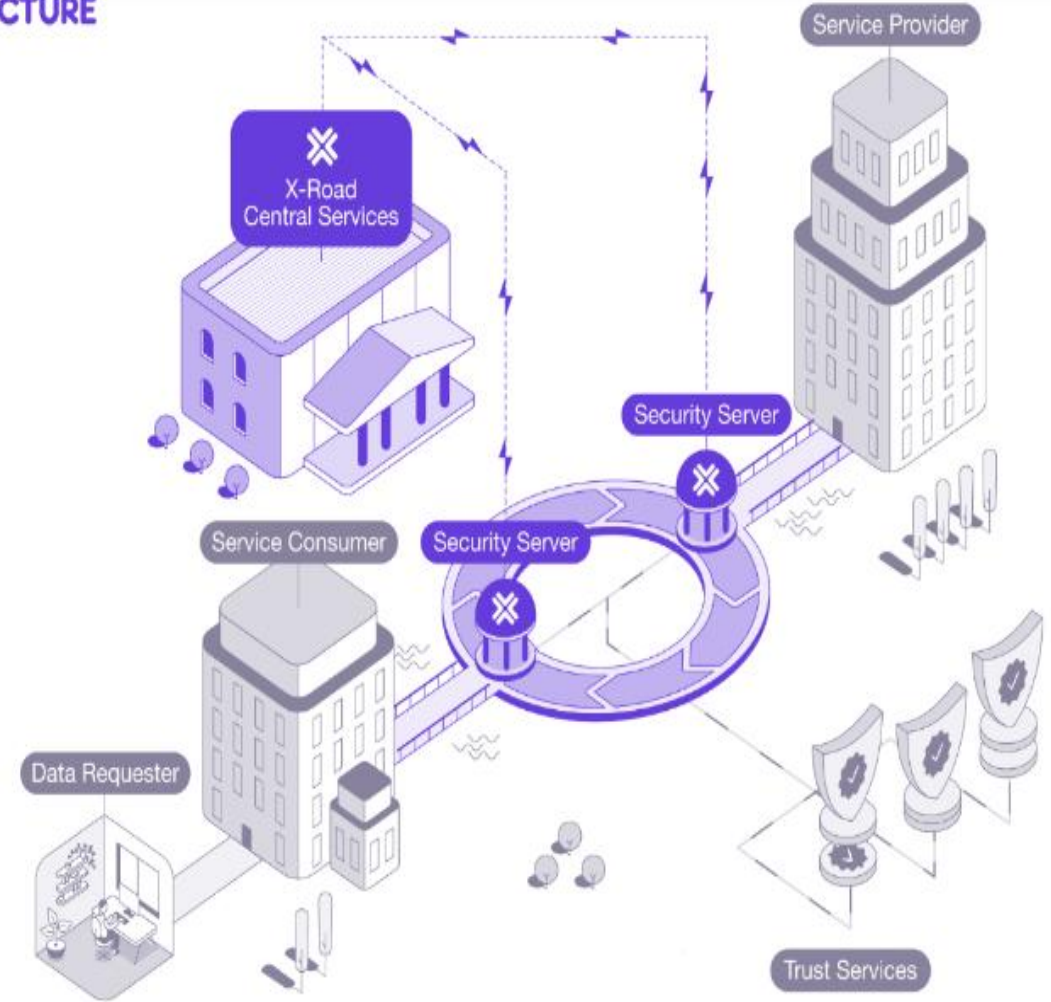
X-ROAD ARCHITECTURE



X-TEE FIVE CORE COMPONENTS (CERTIFICATION AUTHORITY)

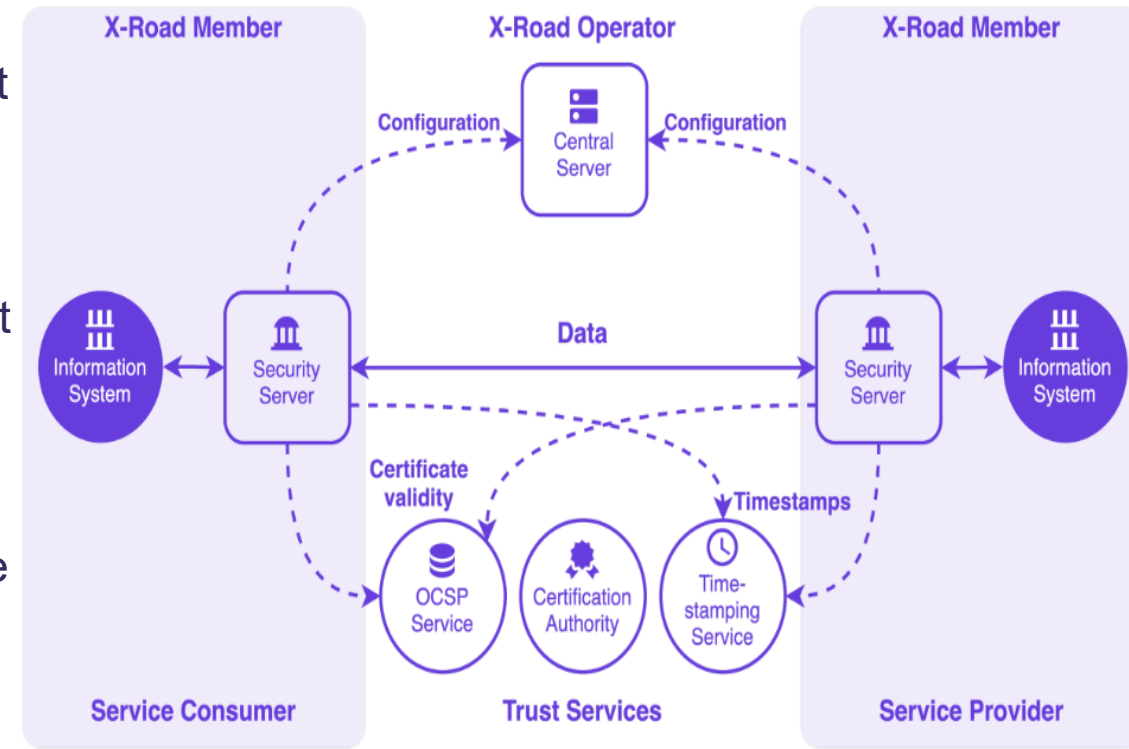
- **Certification Authority (digital certificate provider/ensurer)**
 - The certification authority (CA) issues certificates to Security Servers (authentication certificates) and X-Road member organizations (signing certificates)
 - The Security Server checks the validity of the signing and authentication certificates via the Online Certificate Status Protocol (OCSP).
 - Only Security Servers with valid signing and authentication certificates can exchange messages with other Security Servers.

X-ROAD ARCHITECTURE



DATA EXCHANGE PROCESS (FROM NIIS DOCUMENTATION)

- 1.The service consumer information system initiates a data transaction by sending a request to the service consumer Security Server.
- 2.The service consumer Security Server signs and logs the request and routes it to the service provider Security Server. The connection between the Security Servers is secured using mutual TLS (mTLS).
- 3.The service provider Security Server verifies and logs the request and forwards it to the service provider information system.
- 4.The information system processes the request and returns a response to the service provider Security Server.
- 5.The service provider Security Server signs and logs the response and forwards it to the service consumer Security Server. The connection between the Security Servers is secured using mTLS.
- 6.The service consumer Security Server verifies and logs the response and forwards it to the service consumer information system.
- 7.The service consumer information processes the response.



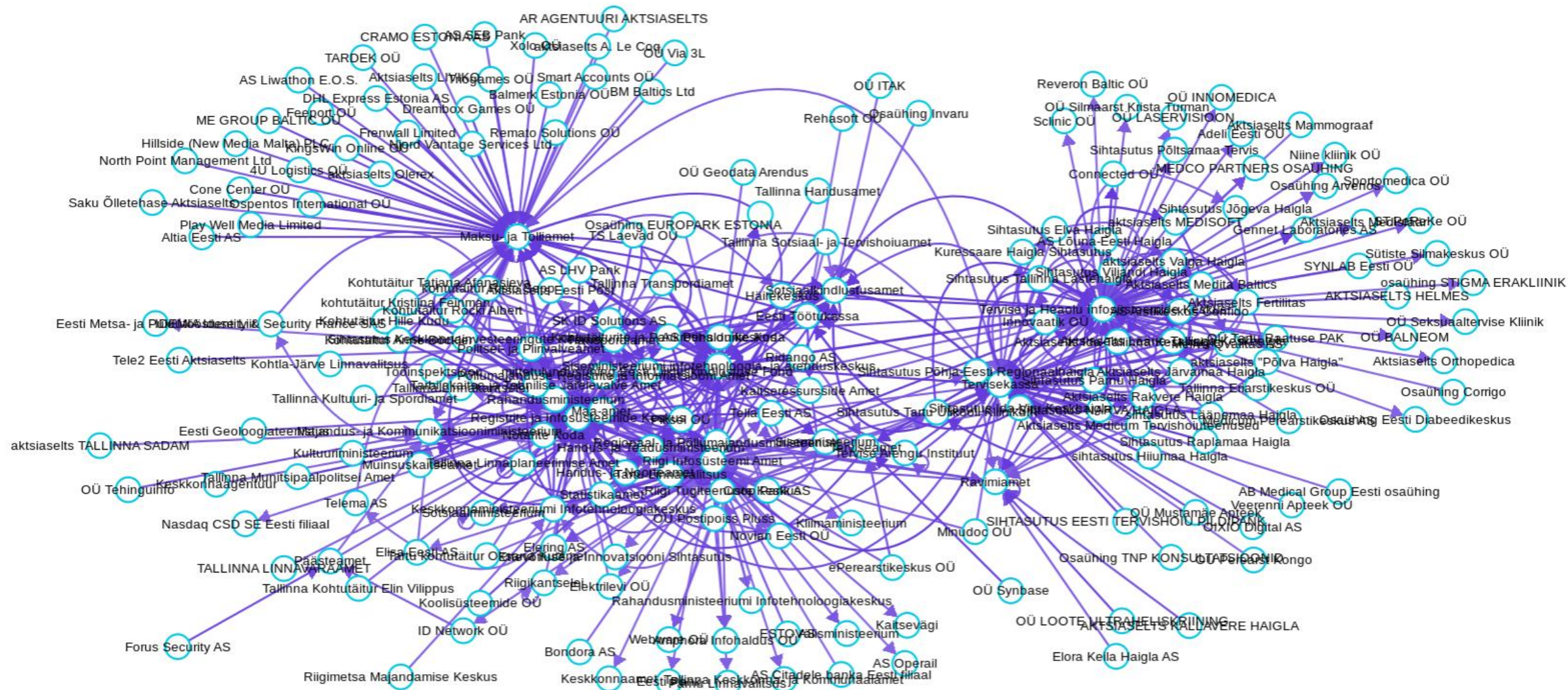
<https://logs.x-tee.ee/visualizer/?profile=EE>

The diagram illustrates a complex network of connections between various entities in Estonia. The central hub is a large blue circle labeled "Maksu- ja Tolliamet". Numerous smaller blue circles are connected to it by lines, representing different organizations, companies, and government departments. The connections are dense, indicating a highly interconnected network.

Entities connected to the central hub include:

- AS G4S Eesti
- Slotbox Limited
- ESTOLIO LIMITED
- Ducks In A Row OÜ
- Vana Lauri OÜ
- OÜ Via 3L
- Ettevõtluse ja Innovatsiooni Sihtasutus
- Kohtutäitur Tatjana Apalnikova
- Kultuuriministeerium
- 1Up Services Ltd
- Blix Gaming Ltd.
- ESTO AS
- Kopikas Entertainment OÜ
- North Point Management Ltd
- Kohtutäitur Anne Böckler
- Xolo OÜ
- Remato Solutions OÜ
- Alta Eesti AS
- Majandus- ja Kommunikatsiooniministeerium
- BM Baltics Ltd
- Kohtutäitur Kristiina Feinman
- Hillside (New Media Malta) PLC
- Siseministeeriumi infotehnoloogia- ja arenduskeskus
- Kohtutäitur Hille Kudu
- TARDEK OÜ
- Far Up Services Ltd.
- Tarbijakaitse ja Tehnilise Järelevalve Amet
- ME GROUP BALTIC OÜ
- Play Well Media Limited
- aktsiaselts Olerex
- aktsiaselts LIVIKO
- aktsiaselts A. Le Coq
- Saku Õllehase Aktsiaselts
- Neste Eesti Aktsiaselts
- CRAMO ESTONIA AS
- Aktsiaselts Olerex Terminal
- AS SEB Pank
- Meliva Qvalitas AS
- Coop Pank AS
- AS LHV Pank
- Smart Accounts OÜ
- AR AGENTUURI AKTSIASELTS
- Happy Hour Solutions Limited
- Tallinna Linnakantselei
- Optiwin OÜ
- AS Lwathon E.O.S.
- Bigbank AS
- Damagi Marketing Solutions Ltd
- Moon Technologies OÜ
- Regionaal- ja Põllumajandusministeerium
- Hitz Gaming OÜ
- Osaühing Tonybet
- Kohtutäitur Rock Albert
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- Tarbijakaitse

X-ROAD FULL ECOSYSTEM DATA EXCHANGE VISUALIZED



CROSS-BORDER DATA EXCHANGE: FINLAND AND ESTONIA



- 2013 Memorandum of cooperation signed between Finland and Estonia for digitalization and cross-border integration
- 2017 Agreement to form a joint non-profit between Finnish and Estonian Ministries of Finance (NIIS)
- 2018 Technical ability released for cross-border data exchange
- 2022 a treaty on the exchange of population data signed between Estonia and Finland

CROSS-BORDER DATA EXCHANGE: FINLAND AND ESTONIA



- When a person moves from Finland to Estonia, they only need to report the change in Estonia, as the information will be automatically updated in the Finnish Population Information System
- Further cooperation: information on the death of non-resident citizens. The exchange of information will gradually extend to address changes, name changes and data on marriages and births.

PERSONAL REFLECTION ON INTEROPERABILITY

- I have conducted interoperability consultations in Morocco and Egypt
- From personal observations, the major problem for internal interoperability is a **lack of defined data quality/standards**, especially when it comes to population or social security registers
 - Duplicates or “Dead Souls”
 - Egypt: 80 million data entries in the social security register – not cleaned or updated information
 - Morocco was interested in X-Road implementation awhile back, but the complexities of the security server installation process
 - Oftentimes, bringing different ministry leaders together in person is the first step towards developing interoperability standards/use cases
 - Business process alignment is essential for establishing interoperability for public service delivery (which means you must know your business processes to begin with!)



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